

**BHAGWAN MAHAVIR UNIVERSITY
FACULTY OF ENGINEERING**



**Object Oriented
Programming using Java
-Practical (1010206202)**

LAB MANUAL

2nd SEMESTER

SUMMER-2024



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LAB MANUAL
Object Oriented Programming using Java
-Practical (1010206202)
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PRACTICAL LIST

Sr. No.	Practical Exercises/Experiments
1	Write a Program that displays Welcome to Java, Learning Java Now and Programming is fun.
2	Write a program that solves the following equation and displays the value x and y: 1) $3.4x + 50.2y = 44.5$ 2) $2.1x + .55y = 5.9$ (Assume Cramer's rule to solve equation $ax + by = e$ $x = \frac{ed - bf}{ad - bc}$ $cx + dy = f$ $y = \frac{af - ec}{ad - bc}$)
3	Write a program that reads a number in meters, converts it to feet, and displays the result.
4	Body Mass Index (BMI) is a measure of health on weight. It can be calculated by taking your weight in kilograms and dividing by the square of your height in meters. Write a program that prompts the user to enter a weight in pounds and height in inches and displays the BMI. Note:- 1 pound = .45359237 Kg and 1 inch = .0254 meters.
5	Write a program that prompts the user to enter three integers and display the integers in decreasing order.
6	Write a program that prompts the user to enter a letter and check whether a letter is a vowel or constant.
7	Assume a vehicle plate number consists of three uppercase letters followed by four digits. Write a program to generate a plate number.
8	Write a program that reads an integer and displays all its smallest factors in increasing order. For example, if input number is 120, the output should be as follows: 2,2,2,3,5.
9	Write the method with following header method. Public static intgcd(int num1, int num2) Write a Program that prompts the user to enter two integers and compute the gcd of two integers.
10	Write a test program that prompts the user to enter ten numbers, invoke a method to reverse the numbers, display the numbers
11	Write a program that generates 6*6 two-dimensional matrix, filled with 0's and 1's, display the matrix, check every row and column have an odd number's of 1's.
12	Write a program that creates a Random object with seed 1000 and displays the first 100 random integers between 1 and 49 using the NextInt (49) method.
13	Write a program for calculator to accept an expression as a string in which the operands and operator are separated by zero or more spaces. For ex: 3+4 and 3 + 4 are acceptable expressions



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14	Write a program that creates an Array List and adds a Loan object, a Date object, a string, and a Circle object to the list, and use a loop to display all elements in the list by invoking the object's toString() method.
15	Write the bin2Dec (string binary String) method to convert a binary string into a decimal number. Implement the bin2Dec method to throw a NumberFormatException if the string is not a binary string.
16	Write a program that prompts the user to enter a decimal number and displays the number in a fraction. Hint: Read the decimal number as a string, extract the integer part and fractional part from the string.
17	Write a program that displays a tic-tac-toe board. A cell may be X, O, or empty. What to display at each cell is randomly decided. The X and O are images in the files X.gif and O.gif.
18	Write a program that moves a circle up, down, left or right using arrow keys.
19	Write a program that displays the color of a circle as red when the mouse button is pressed and as blue when the mouse button is released.
20	Write a GUI program that use button to move the message to the left and right and use the radio button to change the color for the message displayed.
21	Write a program to create a file name 123.txt, if it does not exist. Append a new data to it if it already exists. Write 150 integers created randomly into the file using Text I/O. Integers are separated by space.
22	Write a recursive method that returns the smallest integer in an array. Write a test program that prompts the user to enter an integer and display its product.
23	Write a generic method that returns the minimum elements in a two-dimensional array.
24	Define My Priority Queue class that extends Priority Queue to implement the clone-able interface and implement the clone() method to clone a priority queue.
25	Write a program that reads words from a text file and displays all the non-duplicate words in descending order. The text file is passed as a command-line argument.



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PRACTICAL-1

Write a Program that displays Welcome to Java, Learning Java Now and Programming is fun.

INPUT:

```
public class WelcomeMessage {  
    public static void main(String[] args) {  
        System.out.println("Welcome to Java");  
        System.out.println("Learning Java Now");  
        System.out.println("Programming is fun");  
    }  
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-ja  
Welcome to Java  
Learning Java Now  
Programming is fun  
  
Process finished with exit code 0
```



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PRACTICAL-2

Write a program that solves the following equation and displays the value x and y:

1) $3.4x + 50.2y = 44.5$ 2) $2.1x + .55y = 5.9$

(Assume Cramer's rule to solve equation

$ax + by = e$ $x = (ed - bf) / (ad - bc)$

$cx + dy = f$ $y = (af - ec) / (ad - bc)$).

INPUT:

```
public class EquationSolver {  
    public static void main(String[] args) {  
        // Coefficients of the equations  
        double a = 3.4, b = 50.2, e = 44.5;  
  
        double c = 2.1, d = 0.55, f = 5.9;  
  
        // Determinant  
        double determinant = a * d - b * c;  
  
        // Calculate x and y using Cramer's rule  
        double x = (e * d - b * f) / determinant;  
        double y = (a * f - e * c) / determinant;  
  
        // Display the solution  
        System.out.println("Solution:");  
        System.out.println("x = " + x);  
        System.out.println("y = " + y);  
    }  
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-j  
Solution:  
x = 2.623901496861419  
y = 0.7087397392563978  
  
Process finished with exit code 0
```



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PRACTICAL-3

Write a program that reads a number in meters, converts it to feet, and displays the result.

INPUT:

```
import java.util.Scanner;
```

```
public class MeterToFeetConverter {  
    public static void main(String[] args) {  
        // Create a Scanner object to read input from the user  
        Scanner scanner = new Scanner(System.in);  
  
        // Prompt the user to enter a number in meters  
        System.out.print("Enter a length in meters: ");  
        double meters = scanner.nextDouble();  
  
        // Close the Scanner to avoid resource leak  
        scanner.close();  
  
        // Convert meters to feet  
        double feet = meters * 3.28084;  
  
        // Display the result  
        System.out.println(meters + " meters is equal to " + feet + " feet.");  
    }  
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-ja  
Enter a length in meters: 25  
25.0 meters is equal to 82.021 feet.  
  
Process finished with exit code 0
```



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PRACTICAL-4

Body Mass Index (BMI) is a measure of health on weight. It can be calculated by taking your weight in kilograms and dividing by the square of your height in meters. Write a program that prompts the user to enter a weight in pounds and height in inches and displays the BMI.

Note:- 1 pound=.45359237 Kg and 1 inch=.0254 meters.

INPUT:

```
import java.util.Scanner;
```

```
public class BMICalculator {  
    public static void main(String[] args) {  
        // Create a Scanner object to read input from the user  
        Scanner scanner = new Scanner(System.in);  
  
        // Prompt the user to enter weight in pounds  
        System.out.print("Enter your weight in pounds: ");  
        double weightPounds = scanner.nextDouble();  
  
        // Prompt the user to enter height in inches  
        System.out.print("Enter your height in inches: ");  
        double heightInches = scanner.nextDouble();  
  
        // Close the Scanner to avoid resource leak  
        scanner.close();  
  
        // Convert weight from pounds to kilograms  
        double weightKg = weightPounds * 0.45359237;  
  
        // Convert height from inches to meters  
        double heightMeters = heightInches * 0.0254;  
  
        // Calculate BMI  
        double bmi = weightKg / (heightMeters * heightMeters);  
  
        // Display the BMI  
        System.out.println("Your BMI is: " + bmi);  
    }  
}
```



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OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-ja
Enter your weight in pounds: 165
Enter your height in inches: 70
Your BMI is: 23.674791967441077

Process finished with exit code 0
```



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PRACTICAL-5

Write a program that prompts the user to enter three integers and display the integers in decreasing order.

INPUT:

```
import java.util.Scanner;
```

```
public class DecreasingOrder {  
    public static void main(String[] args) {  
        // Create a Scanner object to read input from the user  
        Scanner scanner = new Scanner(System.in);  
  
        // Prompt the user to enter three integers  
        System.out.print("Enter the first integer: ");  
        int num1 = scanner.nextInt();  
        System.out.print("Enter the second integer: ");  
        int num2 = scanner.nextInt();  
        System.out.print("Enter the third integer: ");  
        int num3 = scanner.nextInt();  
  
        // Close the Scanner to avoid resource leak  
        scanner.close();  
  
        // Find the maximum among the three integers  
        int max = num1;  
        if (num2 > max) {  
            max = num2;  
        }  
        if (num3 > max) {  
            max = num3;  
        }  
  
        // Find the minimum among the three integers  
        int min = num1;  
        if (num2 < min) {  
            min = num2;  
        }  
        if (num3 < min) {  
            min = num3;  
        }  
    }  
}
```



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```
}  
  
// Find the middle number  
int middle = num1 + num2 + num3 - max - min;  
  
// Display the integers in decreasing order  
System.out.println("Integers in decreasing order: " + max + ", " + middle + ", " + min);  
}  
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-java  
Enter the first integer: 25  
Enter the second integer: 13  
Enter the third integer: 67  
Integers in decreasing order: 67, 25, 13  
  
Process finished with exit code 0
```



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PRACTICAL-6

Write a program that prompts the user to enter a letter and check whether a letter is a vowel or constant.

INPUT:

```
import java.util.Scanner;
```

```
public class VowelOrConsonant {  
    public static void main(String[] args) {  
        // Create a Scanner object to read input from the user  
        Scanner scanner = new Scanner(System.in);  
  
        // Prompt the user to enter a letter  
        System.out.print("Enter a letter: ");  
        char letter = scanner.next().charAt(0);  
  
        // Close the Scanner to avoid resource leak  
        scanner.close();  
  
        // Convert the letter to lowercase for case-insensitive comparison  
        letter = Character.toLowerCase(letter);  
  
        // Check if the entered character is a vowel or consonant  
        if (letter >= 'a' && letter <= 'z') {  
            if (letter == 'a' || letter == 'e' || letter == 'i' || letter == 'o' || letter == 'u') {  
                System.out.println(letter + " is a vowel.");  
            } else {  
                System.out.println(letter + " is a consonant.");  
            }  
        } else {  
            System.out.println("Invalid input. Please enter a valid letter from the alphabet.");  
        }  
    }  
}
```



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OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-jav  
Enter a letter: B  
b is a consonant.  
  
Process finished with exit code 0
```

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-java  
Enter a letter: U  
u is a vowel.  
  
Process finished with exit code 0
```



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PRACTICAL-7

Assume a vehicle plate number consists of three uppercase letters followed by four digits. Write a program to generate a plate number.

INPUT:

```
import java.util.Random;
```

```
public class PlateNumberGenerator {  
    public static void main(String[] args) {  
        // Generate three random uppercase letters  
        StringBuilder letters = new StringBuilder();  
        Random random = new Random();  
        for (int i = 0; i < 3; i++) {  
            char letter = (char) ('A' + random.nextInt(26));  
            letters.append(letter);  
        }  
  
        // Generate four random digits  
        StringBuilder digits = new StringBuilder();  
        for (int i = 0; i < 4; i++) {  
            int digit = random.nextInt(10);  
            digits.append(digit);  
        }  
  
        // Concatenate letters and digits to form the plate number  
        String plateNumber = letters.toString() + digits.toString();  
  
        // Display the generated plate number  
        System.out.println("Generated Plate Number: " + plateNumber);  
    }  
}
```



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OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaag  
Generated Plate Number: AJR9628  
  
Process finished with exit code 0
```

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaage  
Generated Plate Number: WZQ6879  
  
Process finished with exit code 0
```



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PRACTICAL-8

Write a program that reads an integer and displays all its smallest factors in increasing order. For example, if input number is 120, the output should be as follows: 2, 2, 2, 3, 5.

INPUT:

```
import java.util.Scanner;
```

```
public class SmallestFactors {  
    public static void main(String[] args) {  
        // Create a Scanner object to read input from the user  
        Scanner scanner = new Scanner(System.in);  
  
        // Prompt the user to enter an integer  
        System.out.print("Enter an integer: ");  
        int number = scanner.nextInt();  
  
        // Close the Scanner to avoid resource leak  
        scanner.close();  
  
        // Display the smallest factors of the input number  
        System.out.print("Smallest factors of " + number + " in increasing order: ");  
        int divisor = 2;  
        while (number > 1) {  
            if (number % divisor == 0) {  
                System.out.print(divisor);  
                number /= divisor;  
                if (number > 1) {  
                    System.out.print(",");  
                }  
            } else {  
                divisor++;  
            }  
        }  
    }  
}
```



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OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagen
Enter an integer: 56
Smallest factors of 56 in increasing order: 2,2,2,7
Process finished with exit code 0
```



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PRACTICAL-9

Write the method with following header method.

Public static intgcd(int num1, int num2)

Write a Program that prompts the user to enter two integers and compute the gcd of two integers

INPUT:

```
import java.util.Scanner;
```

```
public class GCDCalculator {
    public static void main(String[] args) {
        // Create a Scanner object to read input from the user
        Scanner scanner = new Scanner(System.in);

        // Prompt the user to enter two integers
        System.out.print("Enter the first integer: ");

        int num1 = scanner.nextInt();
        System.out.print("Enter the second integer: ");
        int num2 = scanner.nextInt();

        // Close the Scanner to avoid resource leak
        scanner.close();

        // Compute the gcd of the two integers
        int gcdResult = gcd(num1, num2);

        // Display the gcd
        System.out.println("GCD of " + num1 + " and " + num2 + " is: " + gcdResult);
    }

    // Method to compute the gcd of two integers using Euclidean algorithm
    public static int gcd(int num1, int num2) {
        if (num2 == 0) {
            return num1;
        } else {
            return gcd(num2, num1 % num2);
        }
    }
}
```



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```
}  
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-java  
Enter the first integer: 65  
Enter the second integer: 13  
GCD of 65 and 13 is: 13  
  
Process finished with exit code 0
```



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PRACTICAL-10

Write a test program that prompts the user to enter ten numbers, invoke a method to reverse the numbers, display the numbers.

INPUT:

```
import java.util.Scanner;
```

```
public class ReverseNumbers {  
    public static void main(String[] args) {  
        // Create a Scanner object to read input from the user  
        Scanner scanner = new Scanner(System.in);  
  
        // Prompt the user to enter ten numbers  
        int[] numbers = new int[10];  
        for (int i = 0; i < 10; i++) {  
            System.out.print("Enter number " + (i + 1) + ": ");  
            numbers[i] = scanner.nextInt();  
        }  
  
        // Close the Scanner to avoid resource leak  
        scanner.close();  
  
        // Reverse the numbers  
        reverseNumbers(numbers);  
  
        // Display the reversed numbers  
        System.out.println("Reversed numbers:");  
        for (int i = 0; i < 10; i++) {  
            System.out.print(numbers[i] + " ");  
        }  
    }  
}  
  
// Method to reverse an array of numbers  
public static void reverseNumbers(int[] numbers) {  
    int left = 0;  
    int right = numbers.length - 1;  
    while (left < right) {  
        // Swap elements at left and right indices  
        int temp = numbers[left];  
        numbers[left] = numbers[right];  
        numbers[right] = temp;  
        left++;  
        right--;  
    }  
}
```



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```
numbers[right] = temp;
```

```
// Move to the next pair of elements
```

```
left++;
```

```
right--;
```

```
}
```

```
}
```

```
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\P
Enter number 1: 12
Enter number 2: 64
Enter number 3: 98
Enter number 4: 61
Enter number 5: 39
Enter number 6: 72
Enter number 7: 17
Enter number 8: 57
Enter number 9: 49
Enter number 10: 15
Reversed numbers:
15 49 57 17 72 39 61 98 64 12
Process finished with exit code 0
```



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PRACTICAL-11

Write a program that generates 6*6 two-dimensional matrix, filled with 0's and 1's, display the matrix, check every row and column have an odd number's of 1's.

INPUT:

```
import java.util.Random;
```

```
public class MatrixChecker {
    public static void main(String[] args) {
        // Generate the 6x6 matrix
        int[][] matrix = generateMatrix();

        // Display the generated matrix
        System.out.println("Generated Matrix:");
        displayMatrix(matrix);

        // Check if every row and column has an odd number of 1's
        boolean rowsAndColsHaveOddOnes = checkRowsAndColumns(matrix);

        // Display the result
        if (rowsAndColsHaveOddOnes) {
            System.out.println("Every row and column has an odd number of 1's.");
        } else {
            System.out.println("At least one row or column does not have an odd number of 1's.");
        }
    }

    // Method to generate a 6x6 matrix filled with 0's and 1's
    public static int[][] generateMatrix() {
        int[][] matrix = new int[6][6];
        Random random = new Random();

        for (int i = 0; i < 6; i++) {
            for (int j = 0; j < 6; j++) {
                matrix[i][j] = random.nextInt(2); // Generate 0 or 1 randomly
            }
        }
    }
}
```



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```
        return matrix;
    }

    // Method to display a 6x6 matrix
    public static void displayMatrix(int[][] matrix) {
        for (int i = 0; i < 6; i++) {
            for (int j = 0; j < 6; j++) {
                System.out.print(matrix[i][j] + " ");
            }
            System.out.println();
        }
    }

    // Method to check if every row and column has an odd number of 1's
    public static boolean checkRowsAndColumns(int[][] matrix) {
        for (int i = 0; i < 6; i++) {
            int rowCount = 0;
            int colCount = 0;
            for (int j = 0; j < 6; j++) {
                rowCount += matrix[i][j];
                colCount += matrix[j][i];
            }
            if (rowCount % 2 != 1 || colCount % 2 != 1) {
                return false;
            }
        }
        return true;
    }
}
```



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OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Progr  
Generated Matrix:  
0 0 1 0 1 1  
0 1 0 0 1 1  
0 1 0 1 0 0  
1 0 0 1 0 0  
1 1 0 1 0 0  
1 1 1 1 1 0  
At least one row or column does not have an odd number of 1's.  
  
Process finished with exit code 0
```



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PRACTICAL-12

Write a program that creates a Random object with seed 1000 and displays the first 100 random integers between 1 and 49 using the NextInt (49) method.

INPUT:

```
import java.util.Random;
```

```
public class RandomNumberGenerator {  
    public static void main(String[] args) {  
        // Create a Random object with seed 1000  
        Random random = new Random(1000);  
  
        // Display the first 100 random integers between 1 and 49  
  
        System.out.println("First 100 random integers between 1 and 49:");  
        for (int i = 0; i < 100; i++) {  
            int randomNumber = random.nextInt(49) + 1; // Generates a random integer between 1  
and 49  
            System.out.print(randomNumber + " ");  
        }  
    }  
}
```



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OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2023.3.2\lib\idea_rt.jar=50077:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2023.3.2\bin" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -classpath "D:\JAVA Programs\BMU 00PJ Practicals\00PJ-P\out\production\00PJ-P" RandomNumberGenerator
First 100 random integers between 1 and 49:
41 2 18 1 47 47 5 34 10 41 26 14 18 30 31 32 43 36 49 41 12 14 11 1 39 10 1 11 36 11 15 27 35 36 32 44 48 36 3
34 17 49 46 44 6 30 2 36 1 26 29 43 26 3 34 31 19 28 5 29 32 36 10 14 34 13 19 37 40 8 18 32 22 27 48 40 12 41
12 27 49 27 28 33 20 31 27 5 8 41 10 42 9 38 4 35 11 37 5 22
Process finished with exit code 0
```



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PRACTICAL-13

Write a program for calculator to accept an expression as a string in which the operands and operator are separated by zero or more spaces.

For ex: 3+4 and 3 + 4 are acceptable expressions

INPUT:

```
import java.util.Scanner;
```

```
public class SimpleCalculator {  
    public static void main(String[] args) {  
        // Create a Scanner object to read input from the user  
        Scanner scanner = new Scanner(System.in);  
  
        // Prompt the user to enter the expression  
        System.out.print("Enter the expression (operands and operators separated by spaces): ");  
        String expression = scanner.nextLine();  
  
        // Close the Scanner to avoid resource leak  
        scanner.close();  
  
        // Remove any extra spaces from the expression  
        expression = expression.replaceAll("\\s+", "");  
  
        // Parse and evaluate the expression  
        double result = evaluateExpression(expression);  
  
        // Display the result  
        System.out.println("Result: " + result);  
    }  
  
    // Method to evaluate the expression  
    public static double evaluateExpression(String expression) {  
        // Split the expression into operands and operators  
        String[] tokens = expression.split("(?<=[-+*/])|(?=[-+*/])");  
  
        // Convert the first operand to a double  
        double result = Double.parseDouble(tokens[0]);
```



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```
// Iterate through the remaining tokens (operators and operands)
for (int i = 1; i < tokens.length; i += 2) {
    // Get the operator and the next operand
    String operator = tokens[i];
    double operand = Double.parseDouble(tokens[i + 1]);

    // Perform the operation based on the operator
    switch (operator) {
        case "+":
            result += operand;
            break;
        case "-":
            result -= operand;
            break;
        case "*":
            result *= operand;
            break;
        case "/":
            result /= operand;
            break;
        default:
            System.out.println("Invalid operator: " + operator);
            return Double.NaN; // Return NaN (Not a Number) for invalid operators
    }
}
return result;
}
```



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OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ I
Enter the expression (operands and operators separated by spaces): 625 + 125
Result: 750.0

Process finished with exit code 0
```



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PRACTICAL-14

Write a program that creates an Array List and adds a Loan object, a Date object, a string, and a Circle object to the list, and use a loop to display all elements in the list by invoking the object's toString() method.

INPUT:

```
import java.util.ArrayList;
import java.util.Date;

class Loan {
    private double amount;

    public Loan(double amount) {
        this.amount = amount;
    }

    @Override
    public String toString() {
        return "Loan amount: " + amount;
    }
}

class Circle {
    private double radius;

    public Circle(double radius) {
        this.radius = radius;
    }

    @Override
    public String toString() {
        return "Circle with radius: " + radius;
    }
}

public class ObjectArrayListExample {
    public static void main(String[] args) {
        // Create an ArrayList to store various objects
        ArrayList<Object> arrayList = new ArrayList<>();

        // Add objects to the ArrayList
        arrayList.add(new Loan(1000.0)); // Adding a Loan object
        arrayList.add(new Date()); // Adding a Date object
        arrayList.add("Hello, World!"); // Adding a String
```



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```
ArrayList.add(new Circle(5.0)); // Adding a Circle object

// Display all elements in the list using a loop
System.out.println("Elements in the ArrayList:");
for (Object obj : arrayList) {
    System.out.println(obj.toString());
}
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaag
Elements in the ArrayList:
Loan amount: 1000.0
Wed Feb 14 22:00:56 IST 2024
Hello, World!
Circle with radius: 5.0

Process finished with exit code 0
```



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PRACTICAL-15

Write the bin2Dec (string binary String) method to convert a binary string into a decimal number. Implement the bin2Dec method to throw a NumberFormatException if the string is not a binary string.

INPUT:

```
public class BinaryToDecimalConverter {
    public static void main(String[] args) {
        String binaryString = "101010"; // Example binary string
        try {
            int decimalNumber = bin2Dec(binaryString);
            System.out.println("Decimal equivalent of " + binaryString + " is: " + decimalNumber);
        } catch (NumberFormatException e) {
            System.out.println("Invalid binary string: " + e.getMessage());
        }
    }
}

public static int bin2Dec(String binaryString) throws NumberFormatException {
    int decimalNumber = 0;
    // Check if the input string is a valid binary string
    for (int i = 0; i < binaryString.length(); i++) {
        char ch = binaryString.charAt(i);
        if (ch != '0' && ch != '1') {
            throw new NumberFormatException("String is not a binary string");
        }
    }
    // Convert binary string to decimal number
    for (int i = binaryString.length() - 1, j = 0; i >= 0; i--, j++) {
        char ch = binaryString.charAt(i);
        if (ch == '1') {
            decimalNumber += Math.pow(2, j);
        }
    }
    return decimalNumber;
}
```



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OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaa  
Decimal equivalent of 101010 is: 42  
  
Process finished with exit code 0
```



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PRACTICAL-16

Write a program that prompts the user to enter a decimal number and displays the number in a fraction.

Hint: Read the decimal number as a string, extract the integer part and fractional part from the string.

INPUT:

```
import java.util.Scanner;
```

```
public class DecimalToFraction {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Prompt the user to enter a decimal number
        System.out.print("Enter a decimal number: ");
        String decimalStr = scanner.nextLine();

        // Extract integer part and fractional part
        String[] parts = decimalStr.split("\\.");
        int integerPart = Integer.parseInt(parts[0]);
        int fractionalPart = Integer.parseInt(parts[1]);

        // Find the denominator based on the number of digits in the fractional part
        int denominator = (int) Math.pow(10, parts[1].length());

        // Simplify the fraction
        int gcd = gcd(fractionalPart, denominator);
        fractionalPart /= gcd;
        denominator /= gcd;

        // Display the number in fraction form
        System.out.println("Fractional representation:");
        System.out.println(integerPart + " " + fractionalPart + "/" + denominator);

        scanner.close();
    }

    // Method to find the greatest common divisor (gcd) of two numbers
    public static int gcd(int a, int b) {
        if (b == 0)
            return a;
        return gcd(b, a % b);
    }
}
```



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OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-ja
Enter a decimal number: 10.63
Fractional representation:
10 63/100

Process finished with exit code 0
```



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PRACTICAL-17

Write a program that displays a tic-tac-toe board. A cell may be X, O, or empty. What to display at each cell is randomly decided. The X and O are images in the files X.gif and O.gif.

INPUT:

```
import javax.swing.*;
import java.awt.*;
import java.util.Random;
```

```
public class TicTacToeBoard extends JFrame {
    private final ImageIcon X_ICON = new ImageIcon("X.gif");
    private final ImageIcon O_ICON = new ImageIcon("O.gif");
    private final ImageIcon EMPTY_ICON = new ImageIcon("Empty.gif");

    private final int ROWS = 3;
    private final int COLS = 3;

    private JLabel[][] cells;

    public TicTacToeBoard() {
        setTitle("Tic-Tac-Toe Board");
        setSize(300, 300);

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // Create the grid layout for the board
        setLayout(new GridLayout(ROWS, COLS));

        // Initialize the cells array
        cells = new JLabel[ROWS][COLS];

        // Populate the board with random X, O, or empty cells
        Random random = new Random();
        for (int row = 0; row < ROWS; row++) {
            for (int col = 0; col < COLS; col++) {
                int randomNumber = random.nextInt(3);
                if (randomNumber == 0) {
                    cells[row][col] = new JLabel(X_ICON);
                } else if (randomNumber == 1) {
                    cells[row][col] = new JLabel(O_ICON);
                } else {
                    cells[row][col] = new JLabel(EMPTY_ICON);
                }
            }
        }
    }
}
```



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```
        } else if (randomNumber == 1) {
            cells[row][col] = new JLabel(O_ICON);
        } else {
            cells[row][col] = new JLabel(EMPTY_ICON);
        }
        add(cells[row][col]);
    }
}

setVisible(true);
}

public static void main(String[] args) {
    new TicTacToeBoard();
}
}
```

OUTPUT:



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PRACTICAL-18

Write a program that moves a circle up, down, left or right using arrow keys.

INPUT:

```
import javax.swing.*;  
import java.awt.*;  
import java.awt.event.*;
```

```
public class MoveCircle extends JFrame {  
    private int circleX = 150; // Initial x-coordinate of the circle  
    private int circleY = 150; // Initial y-coordinate of the circle  
  
    private final int CIRCLE_RADIUS = 50; // Radius of the circle  
  
    public MoveCircle() {  
        setTitle("Move Circle");  
        setSize(400, 400);  
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
  
        JPanel panel = new JPanel() {  
            @Override  
            protected void paintComponent(Graphics g) {  
                super.paintComponent(g);  
                // Draw the circle  
                g.setColor(Color.RED);  
                g.fillOval(circleX - CIRCLE_RADIUS, circleY - CIRCLE_RADIUS, 2 *  
CIRCLE_RADIUS, 2 * CIRCLE_RADIUS);  
            }  
        };  
  
        panel.setFocusable(true);  
        panel.requestFocusInWindow();  
  
        // Add KeyListener to the panel  
        panel.addKeyListener(new KeyAdapter() {  
            @Override  
            public void keyPressed(KeyEvent e) {  
                int keyCode = e.getKeyCode();  
                switch (keyCode) {  
                    case KeyEvent.VK_UP:
```



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```
        circleY -= 10;
        break;
    case KeyEvent.VK_DOWN:
        circleY += 10;
        break;
    case KeyEvent.VK_LEFT:
        circleX -= 10;
        break;
    case KeyEvent.VK_RIGHT:
        circleX += 10;
        break;
    }
    panel.repaint();
}
});

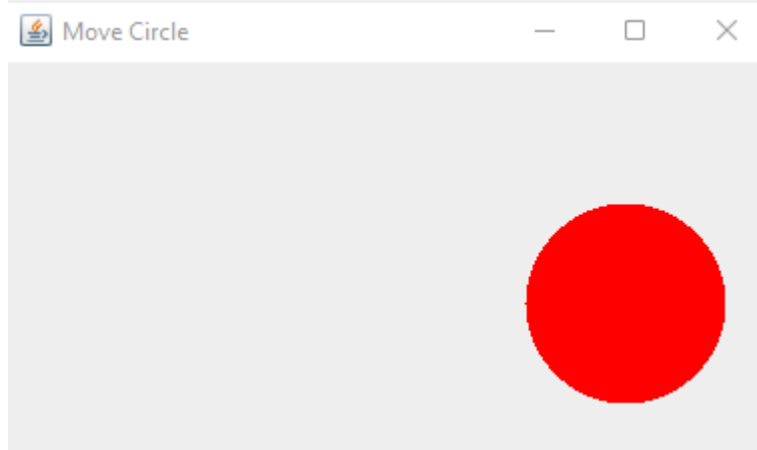
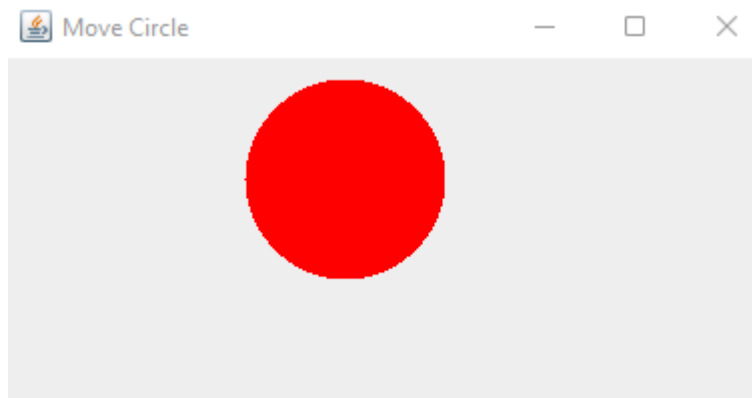
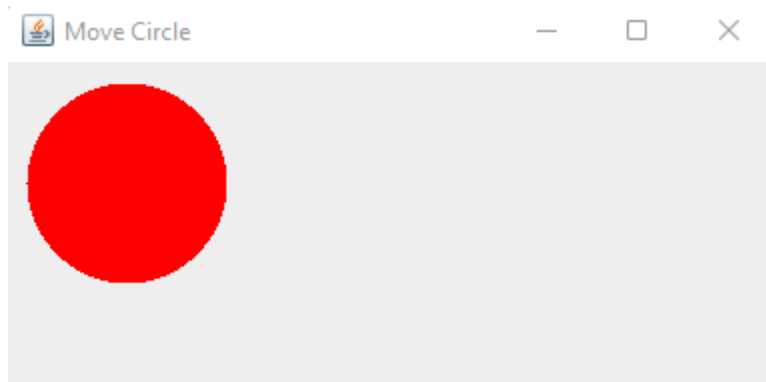
add(panel);
setVisible(true);
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(MoveCircle::new);
}
}
```



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OUTPUT:





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PRACTICAL-19

Write a program that displays the color of a circle as red when the mouse button is pressed and as blue when the mouse button is released.

INPUT:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
```

```
public class CircleColorChange extends JFrame {
    private int circleX = 150; // Initial x-coordinate of the circle
    private int circleY = 150; // Initial y-coordinate of the circle
    private final int CIRCLE_RADIUS = 50; // Radius of the circle
    private Color circleColor = Color.BLUE; // Initial color of the circle

    public CircleColorChange() {
        setTitle("Circle Color Change");
        setSize(400, 400);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        JPanel panel = new JPanel() {
            @Override
            protected void paintComponent(Graphics g) {
                super.paintComponent(g);
                // Draw the circle with the current color
                g.setColor(circleColor);
                g.fillOval(circleX - CIRCLE_RADIUS, circleY - CIRCLE_RADIUS, 2 *
CIRCLE_RADIUS, 2 * CIRCLE_RADIUS);
            }
        };

        panel.addMouseListener(new MouseAdapter() {
            @Override
            public void mousePressed(MouseEvent e) {
                circleColor = Color.RED;
                panel.repaint();
            }
        });
    }
}
```



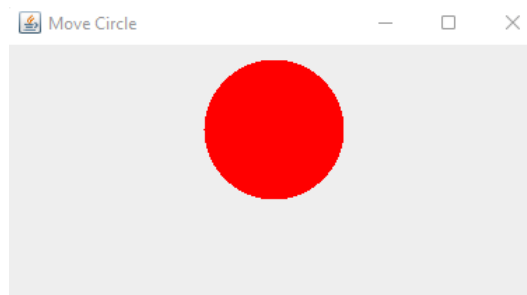
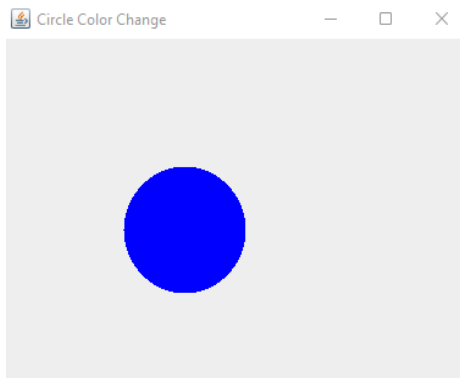
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```
@Override
public void mouseReleased(MouseEvent e) {
    circleColor = Color.BLUE;
    panel.repaint();
}

add(panel);
setVisible(true);
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(CircleColorChange::new);
}
}
```

OUTPUT:





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PRACTICAL-20

Write a GUI program that use button to move the message to the left and right and use the radio button to change the color for the message displayed.

INPUT:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
```

```
public class MessageApp extends JFrame {
    private JLabel messageLabel;
    private JButton leftButton;
    private JButton rightButton;
    private JRadioButton redRadioButton;
    private JRadioButton greenRadioButton;
    private JRadioButton blueRadioButton;
    private ButtonGroup colorGroup;

    public MessageApp() {
        setTitle("Message App");
        setSize(400, 200);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // Create components
        messageLabel = new JLabel("Hello, Swing!", JLabel.CENTER);
        leftButton = new JButton("Left");
        rightButton = new JButton("Right");

        redRadioButton = new JRadioButton("Red");
        greenRadioButton = new JRadioButton("Green");
        blueRadioButton = new JRadioButton("Blue");

        colorGroup = new ButtonGroup();
        colorGroup.add(redRadioButton);
        colorGroup.add(greenRadioButton);
        colorGroup.add(blueRadioButton);

        // Set layout
        setLayout(new BorderLayout());
```



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```
// Add components to the frame
JPanel buttonPanel = new JPanel();
buttonPanel.add(leftButton);
buttonPanel.add(rightButton);
add(buttonPanel, BorderLayout.NORTH);

JPanel radioPanel = new JPanel();
radioPanel.add(redRadioButton);
radioPanel.add(greenRadioButton);
radioPanel.add(blueRadioButton);
add(radioPanel, BorderLayout.SOUTH);

add(messageLabel, BorderLayout.CENTER);

// Register listeners
leftButton.addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        moveMessage(-10);
    }
});

rightButton.addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        moveMessage(10);
    }
});

ActionListener colorListener = new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        if (redRadioButton.isSelected()) {
            messageLabel.setForeground(Color.RED);
        } else if (greenRadioButton.isSelected()) {
            messageLabel.setForeground(Color.GREEN);
        } else if (blueRadioButton.isSelected()) {
            messageLabel.setForeground(Color.BLUE);
        }
    }
}
```



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```
    }  
};  
  
redRadioButton.addActionListener(colorListener);  
greenRadioButton.addActionListener(colorListener);  
blueRadioButton.addActionListener(colorListener);  
  
setVisible(true);  
}  
  
private void moveMessage(int offset) {  
    int x = messageLabel.getLocation().x + offset;  
    int y = messageLabel.getLocation().y;  
    messageLabel.setLocation(x, y);  
}  
  
public static void main(String[] args) {  
    SwingUtilities.invokeLater(new Runnable() {  
        @Override  
        public void run() {  
            new MessageApp();  
        }  
    });  
}  
}
```



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OUTPUT:





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PRACTICAL-21

Write a program to create a file name 123.txt, if it does not exist. Append a new data to it if it already exists. Write 150 integers created randomly into the file using Text I/O. Integers are separated by space

INPUT:

```
import java.io.BufferedWriter;  
import java.io.File;  
import java.io.FileWriter;  
import java.io.IOException;  
import java.util.Random;
```

```
public class WriteToFile {  
    public static void main(String[] args) {  
        String fileName = "123.txt";  
        int numberOfIntegers = 150;  
  
        // Create a File object  
        File file = new File(fileName);  
  
        try {  
            // Create the file if it does not exist  
            if (!file.exists()) {  
                file.createNewFile();  
            }  
  
            // Open the file in append mode  
            FileWriter fileWriter = new FileWriter(file, true);  
            BufferedWriter bufferedWriter = new BufferedWriter(fileWriter);  
  
            // Generate and write random integers to the file  
            Random random = new Random();  
            for (int i = 0; i < numberOfIntegers; i++) {  
                int randomNumber = random.nextInt(1000); // Generate a random integer between 0  
and 999  
                bufferedWriter.write(randomNumber + " ");  
            }  
  
            // Close the BufferedWriter  
            bufferedWriter.close();  
        }  
    }  
}
```



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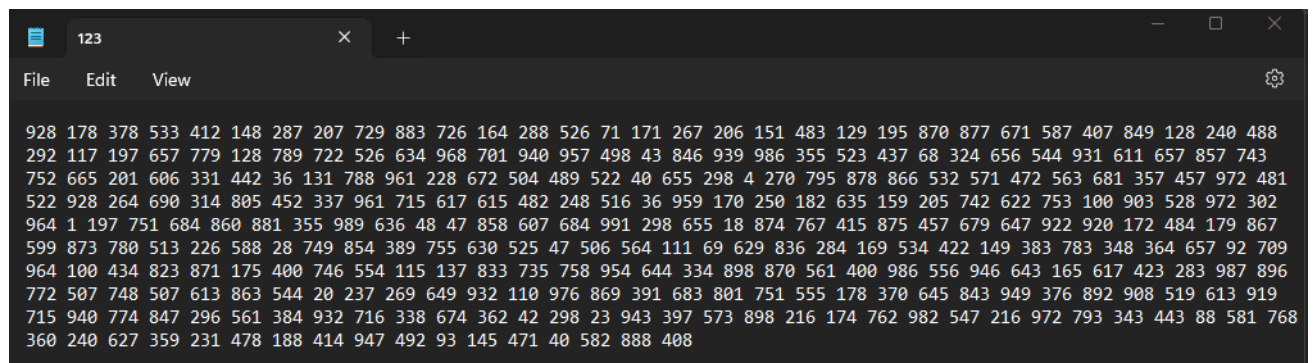
```
System.out.println("Data has been written to the file.");

    } catch (IOException e) {
        System.err.println("Error occurred while writing to the file: " + e.getMessage());
    }
}
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent
Data has been written to the file.

Process finished with exit code 0
```





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PRACTICAL-22

Write a recursive method that returns the smallest integer in an array. Write a test program that prompts the user to enter an integer and display its product.

INPUT:

```
import java.util.Scanner;
```

```
public class SmallestInteger {
    public static void main(String[] args) {
        // Test the findSmallest method
        int[] array = {5, 3, 8, 2, 9, 1};
        int smallest = findSmallest(array);
        System.out.println("The smallest integer in the array is: " + smallest);

        // Test the productPrompt method
        productPrompt();
    }

    // Recursive method to find the smallest integer in an array
    public static int findSmallest(int[] array) {
        return findSmallestHelper(array, array.length - 1, array[0]);
    }

    private static int findSmallestHelper(int[] array, int index, int smallestSoFar) {
        if (index < 0) {
            return smallestSoFar;
        }
        return findSmallestHelper(array, index - 1, Math.min(smallestSoFar, array[index]));
    }

    // Method to prompt the user to enter integers and display their product
    public static void productPrompt() {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the number of integers: ");
        int n = scanner.nextInt();
        int[] numbers = new int[n];
```



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```
System.out.println("Enter " + n + " integers:");
for (int i = 0; i < n; i++) {
    numbers[i] = scanner.nextInt();
}

int product = getProduct(numbers);
System.out.println("Product of the integers entered: " + product);

scanner.close();
}

// Method to calculate the product of integers in an array
public static int getProduct(int[] numbers) {
    int product = 1;
    for (int num : numbers) {
        product *= num;
    }
    return product;
}
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagen
The smallest integer in the array is: 1
Enter the number of integers: 9
Enter 9 integers:
6
7
12
5
18
20
6
7
8
Product of the integers entered: 304819200
Process finished with exit code 0
```



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PRACTICAL-23

Write a generic method that returns the minimum elements in a two-dimensional array.

INPUT:

```
public class MinTwoDArray {

    public static void main(String[] args) {
        Integer[][] array1 = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
        Double[][] array2 = {{1.5, 2.5, 3.5}, {4.5, 5.5, 6.5}, {7.5, 8.5, 9.5}};

        System.out.println("Minimum element in array1: " + findMin(array1));
        System.out.println("Minimum element in array2: " + findMin(array2));
    }

    public static <T extends Comparable<T>> T findMin(T[][] array) {
        if (array == null || array.length == 0 || array[0].length == 0) {
            throw new IllegalArgumentException("Array is empty or null");
        }

        T min = array[0][0];
        for (T[] row : array) {
            for (T element : row) {
                if (element.compareTo(min) < 0) {
                    min = element;
                }
            }
        }
        return min;
    }
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaage
Minimum element in array1: 1
Minimum element in array2: 1.5

Process finished with exit code 0
```



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PRACTICAL-24

Define My Priority Queue class that extends Priority Queue to implement the clone-able interface and implement the clone() method to clone a priority queue

INPUT:

```
import java.util.PriorityQueue;

public class MyPriorityQueue<E> extends PriorityQueue<E> implements Cloneable {

    @Override
    public MyPriorityQueue<E> clone() {
        MyPriorityQueue<E> clonedQueue = new MyPriorityQueue<>();
        clonedQueue.addAll(this);
        return clonedQueue;
    }

    public static void main(String[] args) {
        // Create an instance of MyPriorityQueue
        MyPriorityQueue<Integer> originalQueue = new MyPriorityQueue<>();

        // Add elements to the original queue
        originalQueue.add(3);
        originalQueue.add(1);
        originalQueue.add(5);
        originalQueue.add(2);

        // Clone the original queue
        MyPriorityQueue<Integer> clonedQueue = originalQueue.clone();

        // Display the original and cloned queues
        System.out.println("Original Queue: " + originalQueue);
        System.out.println("Cloned Queue: " + clonedQueue);
    }
}
```



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OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagen  
Original Queue: [1, 2, 5, 3]  
Cloned Queue: [1, 2, 5, 3]  
  
Process finished with exit code 0
```



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PRACTICAL-25

Write a program that reads words from a text file and displays all the non-duplicate words in descending order. The text file is passed as a command-line argument.

INPUT:

```
import java.io.File;
import java.io.FileNotFoundException;
import java.util.*;

public class NonDuplicateWords {
    public static void main(String[] args) {
        // Check if a filename is provided as a command-line argument
        if (args.length != 1) {
            System.out.println("Usage: java NonDuplicateWords <filename>");
            return;
        }

        String filename = args[0];
        File file = new File(filename);

        // Check if the file exists
        if (!file.exists() || !file.isFile()) {
            System.out.println("File not found: " + filename);
            return;
        }

        // Read words from the file and store them in a TreeSet to remove duplicates and sort them
        TreeSet<String> uniqueWords = new TreeSet<>(Collections.reverseOrder());
        try (Scanner scanner = new Scanner(file)) {
            while (scanner.hasNext()) {
                String word = scanner.next().toLowerCase(); // Convert to lowercase for
                case-insensitive comparison
                uniqueWords.add(word);
            }
        } catch (FileNotFoundException e) {
            System.out.println("Error reading file: " + e.getMessage());
            return;
        }
    }
}
```



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```
}  
  
    // Display non-duplicate words in descending order  
    System.out.println("Non-duplicate words in descending order:");  
    for (String word : uniqueWords) {  
        System.out.println(word);  
    }  
}  
}
```

OUTPUT: